CSCI4146/6409 Process of Data Science
Project 2: Explainable AI (XAI) Professional Model Audit

Due Date: June 17th, 2026

This project is designed as a professional model-audit engagement, not a leaderboard contest. You are hired as an external audit team to evaluate a credit-risk model that appears strong overall, but may fail for specific cases and subgroups. Your goal is to produce decision-ready evidence about model behavior, risk concentration, and practical remediation.

Project logistics:

- Programming language: Python (Jupyter Notebook environment);
- Primary dataset: LendingClub loan data (provided in project 2/data);
- Due date: posted in Brightspace/course syllabus;
- Team policy: follow current course policy (individual or team-based as announced).

Marking scheme and requirements: Full marks are given for (1) working, readable, reasonably efficient, documented analysis code that addresses the audit goals, and (2) clear, defensible interpretations that connect technical evidence to decision implications. High-level claims without quantitative support will lose marks.

Please follow the course collaboration and integrity policy. If you use external resources (including AI tools), you must acknowledge them and remain fully responsible for correctness, interpretation, and final arguments.

What/how to submit your work: Submission package: (a) completed workbook notebook `project_workbook.ipynb`; (b) notebook with outputs visible and PDF export; (c) concise audit memo (1–2 pages, separate PDF or clearly labeled notebook section); (d) 10-minute presentation slides summarizing framing, evidence, decisions, and limitations; (e) source and AI-usage disclosure (if applicable).

1 Before You Start

In the tutorial notebook and workbook, you are expected to practice the full audit pipeline: black-box performance evaluation, local error explanation, global interaction analysis, subgroup slicing diagnostics, and remediation design.

This project emphasizes explanation quality and audit reasoning. Strong model metrics alone are insufficient.

Frequently asked clarifications

- *Do we need the highest possible AUC?* No. You need a credible model and a strong audit; a slightly weaker model with better audit quality can score higher than a stronger model with weak analysis.
- *Can we use methods other than SHAP/LIME?* Yes, but SHAP and LIME evidence is mandatory for required sections; extra methods are optional and must be justified.
- *What counts as a good local explanation section?* Not just plots; explain why highlighted contributions are plausible, risky, or suspicious for each case.
- *What if local and global explanations disagree?* This is acceptable; discuss local context, nonlinearity, interactions, and sample effects.
- *Do we have to fix fairness completely?* No; identify subgroup risk and propose feasible mitigation with trade-offs and limitations.

2 Main Assignment

You are consulting as an XAI audit team. Please answer all required tasks in your workbook notebook and link each major claim to computed evidence. The task structure below aligns directly with the course-wide rubric categories.

R1. Problem Framing & Metrics (15%)

Define the business objective and success criteria before modeling.

Minimum requirements:

- State a clear decision objective and stakeholder.
- Define a metric hierarchy (north-star, input, and guardrail metrics).
- Justify threshold/capacity assumptions and what success means operationally.
- Explain why this metric design is appropriate for credit-risk auditing.

R2. Technical Execution (25%)

Implement a correct and reproducible end-to-end modeling and explanation pipeline.

Minimum requirements:

- Build a baseline model with train/validation/test discipline and leakage controls.

- Report ROC-AUC, F1, precision, recall, confusion matrix, and threshold table.
- Provide local explanation evidence for at least three high-value FP/FN cases.
- Provide global explanation evidence (importance + PDP/effect view + at least one interaction).
- Save and document model artifacts for reproducibility.

R3. Insight & Decision Quality (20%)

Turn model outputs into actionable decisions.
Minimum requirements:

- Identify major error patterns and likely business impact.
- Connect explanation findings to at least two concrete recommendations.
- Quantify trade-offs (for example precision-recall, volume-capacity, false positive/negative impact).
- Prioritize which actions should be implemented first and justify why.

R4. Debugging & Validation (15%)

Demonstrate robustness, error analysis, and limitation awareness.
Minimum requirements:

- Perform explicit leakage checks and document assumptions.
- Include subgroup/slice diagnostics and discuss potential bias or uneven risk.
- Compare at least one alternative setup (for example threshold or feature-set change) and discuss stability.
- Clearly state limitations and failure modes.

R5. Communication (Report + Talk) (15%)

Present your work clearly in both written and spoken forms.
Minimum requirements:

- Keep notebook/report narrative structured, concise, and evidence-driven.
- In the 10-minute talk, highlight framing, method, key findings, and recommendations.
- Ensure visuals are readable and prioritized by decision importance.
- Make confidence and uncertainty explicit.

R6. Q&A / Technical Defense (10%)

Be prepared to defend your choices and interpretations.
Minimum requirements:

- Justify method choices, assumptions, and threshold decisions when asked.
- Explain at least one model error case live and why the model likely failed.
- Defend subgroup-risk claims with concrete metrics.
- Inability to justify major decisions/results may lead to deductions.

Professional Writing Requirement (Mandatory)

Your notebook must read like an audit report, not only a code dump. For each milestone, include: (a) Claim, (b) Evidence, (c) Interpretation, (d) Decision implication, and (e) Limitation.

3 Required Artifacts (Rubric-Aligned)

Rubric area	Required outputs	Minimum evidence quality
R1	Problem statement, stakeholder context, metric hierarchy, success criteria	Metrics map clearly connected to decision objective
R2	Model pipeline, evaluation outputs, local/global XAI evidence, saved artifacts	Correct implementation, reproducibility, and method justification
R3	Recommendation set with impact and trade-off estimates	Evidence-backed prioritization and actionable decisions
R4	Leakage checks, slice diagnostics, robustness comparison, limitation notes	Transparent validation and realistic risk discussion
R5/R6	Audit memo + slides + defense readiness notes	Clear communication and ability to justify technical choices

4 Marking Scheme

Category	Weight	Description
Problem Framing & Metrics	15%	Clear definition of business objective; well-structured metric hierarchy (north-star, input, guardrail); and justified success criteria
Technical Execution	25%	Correct and effective use of data processing, modeling, and analytical methods; appropriate choice of tools and techniques
Insight & Decision Quality	20%	Ability to generate actionable insights and clearly connect results to decisions or measurable business impact
Debugging & Validation	15%	Thorough error analysis, robustness checks, bias detection, and clear discussion of assumptions and limitations
Communication (Report + Talk)	15%	Clear, well-structured report and effective 10-minute presentation; strong prioritization and explanation of key findings
Q&A / Technical Defense	10%	Ability to answer questions from peers and instructor; demonstrates understanding of methods, assumptions, and limitations; inability to justify decisions or results will result in deduction

High marks require strength across all components. A technically correct notebook with weak interpretation will not receive top marks.

5 AI/LLM Usage and Integrity

AI tools may be used as support tools, but not as replacements for your reasoning. If used, you must disclose usage and verify all generated content. Any submission that cannot be explained by the student(s) may be penalized and referred under course integrity policy.

6 Final Submission Checklist

Before submission, ensure all items are satisfied:

- Notebook runs end-to-end with visible outputs.
- All rubric-aligned tasks (R1–R6) are addressed with code and interpretation.
- Leakage risks are explicitly addressed.
- At least three local cases are explained with evidence.
- At least one interaction effect is identified and interpreted.
- Slice analysis includes subgroup metrics and risk gap discussion.
- Mitigation plan includes feasibility, trade-off, and validation notes.
- Presentation slides are ready for a 10-minute talk.
- Team is prepared for technical Q&A defense.
- Audit memo and AI/source disclosures are included (if applicable).

End Note

The educational goal of this project is to practice trustworthy data science: building useful models, auditing behavior rigorously, and communicating responsible decisions under uncertainty.